4.4.1 Typical power ratings of appliances

At the textbook's rated 220 V, example powers are 2000 W for a heater, 1200 W for an oven, 1000 W for an air conditioner, 120 W for a refrigerator and 100 W for a television. These illustrate orders of magnitude, not universal constants. Use an actual appliance's rating or the question's supplied data.

4.4.2 Live, neutral and earth wires

In the textbook mains model, neutral is approximately at earth potential while live alternates between positive and negative potentials. This provides an alternating voltage between them. Operating current uses live and neutral; earth mainly provides a protective fault path.

The switch and fuse are placed in the live wire. Opening them disconnects the appliance from live. Opening neutral alone can leave internal parts connected to live.

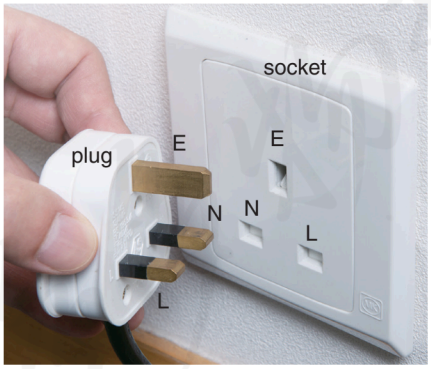


Fig 4.2b *Three-pin plug and socket.*

L = live, N = neutral, E = earth. Facing the socket: E at the top, N on the left and L on the right. Match each plug pin to the socket hole with the same letter.

4.4.3 Household electrical protection

Different protections respond to different faults. A fuse or overcurrent circuit breaker disconnects live when current is excessive, reducing wire overheating. Its rating should be just above normal operating current. A harmful current through a person may be below the fuse rating, so a fuse alone may not prevent electric shock.

If live touches an earthed metal case, the earth wire provides a low-resistance fault path that helps protection disconnect the supply. Double insulation uses two insulating barriers to separate live parts from the user.

Device	Trigger	Main function
Fuse / overcurrent circuit breaker	Current exceeds its limit	Reduces overheating of wires or appliances
Residual-current circuit breaker (RCCB)	A sufficient difference between live and neutral currents	Detects current escaping by another path and disconnects the supply rapidly

4.4.4 Direct and alternating current

Direct current keeps the same direction although its magnitude may vary. Alternating current reverses periodically, with corresponding voltage polarity changes. Inspect whether current crosses zero and reverses, rather than merely whether it fluctuates. A pulsating current that never reverses is still direct current.

R.m.s. values and alternating-current power

R.m.s. current or voltage is the equivalent d.c. value producing the same average heating power in the same fixed resistance. For a purely resistive load, voltage and current are in phase, so their r.m.s. product gives average power. Zero mean current over a cycle does not mean zero energy dissipation.

$$P = I_{\text{rms}}^2 R = \frac{V_{\text{rms}}^2}{R} = V_{\text{rms}} I_{\text{rms}}$$

Pure constant resistance R (Ω), so voltage and current are in phase. Average power $\bar{P}$: W; RMS voltage: V; RMS current: A.

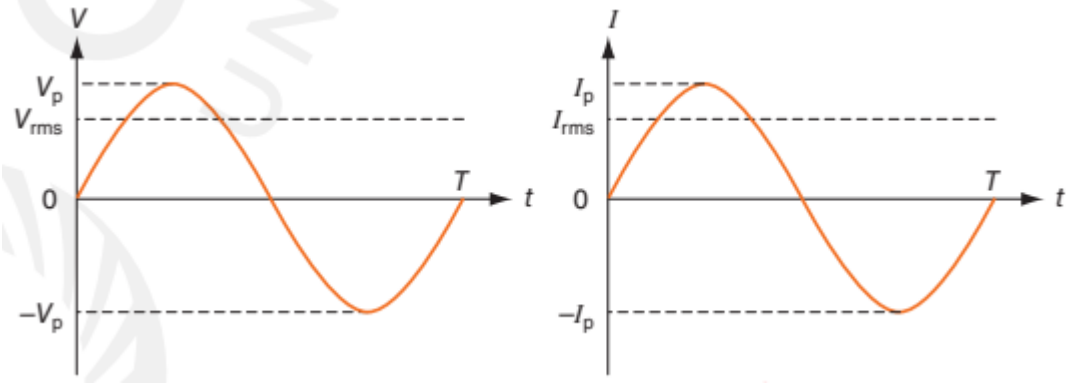
Peak and r.m.s. values of a sinusoid

For a sinusoid with no d.c. offset, the r.m.s. value equals peak value divided by the square root of two. Peak is measured from zero to the maximum magnitude. Peak-to-peak is twice the peak value. This conversion is waveform-specific: square waves and other shapes require their own r.m.s. calculation.

$$V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}}$$

$$I_{\text{rms}} = \frac{I_{\text{peak}}}{\sqrt{2}}$$

Only for sinusoidal voltage/current with zero DC offset. Peak means amplitude, not peak-to-peak.



Zero lines, peak values and periods of sinusoidal a.c.; r.m.s. value is not peak-to-peak value.